Gioia Zheng

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RESEARCH INTERESTS

Machine learning, information retrieval and natural language processing, with an interest in evaluating learning systems through controlled experiments, interpretable diagnostics and reproducible software.

EDUCATION

TU Darmstadt Winter 2026

M.Sc. Computer Science

IAS Robot Learning project: Learning Amphibious Legged Locomotion (ongoing). Prof. Jan Peters.

Sapienza University of Rome Winter 2022

B.Sc. Applied Computer Science and Artificial Intelligence

University of Eastern Finland Aug – Sep 2026

Visiting student · Artificial Intelligence for Computer Games · 5 ECTS, Pass

RESEARCH AND PROFESSIONAL EXPERIENCE

Undergraduate Thesis Research

Sapienza University of Rome · Supervisor: Prof. Marco Raoul Marini

- Studying retrieval and reranking across datasets, focusing on evidence coverage and answer quality.
- Conducted controlled comparisons and query-level error analysis; developed reproducible evaluation workflows.

NLP Algorithm Engineer Intern

Feb – May 2026

Microsoft · Remote

- Implemented retrieval, reranking and question-answering components in Python; developed evaluation and result-validation workflows.

Web Development Intern

Mar – Jun 2026

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- Maintained production web systems and automated internal workflows in a client-facing environment.

SELECTED PROJECTS

MS MARCO Generative QA · GitHub
Modular retrieval and generation pipeline with statistical comparisons, grounding checks and reproducible experiment records.

RAG Observatory · GitHub
Trace-based toolkit for inspecting retrieved evidence, generated answers and failure patterns in retrieval-augmented systems.

Q-learning Exploitability · GitHub
Tabular reinforcement learning with exact best-response evaluation, opponent mixtures and symmetry-aware representations.

OPEN SOURCE AND SKILLS

Contributions to MTEB, Pyserini and Open RAG Eval, including evaluation correctness, dataset maintenance and reproducibility checks.

Programming and tools Python, SQL, Git, Linux, Docker, pytest, GitHub Actions.
Machine learning PyTorch, Transformers, scikit-learn; retrieval, NLP and reinforcement-learning evaluation.
Languages Chinese (native), Italian (bilingual), English (professional), German (beginner).